

Netflix Data Analysis

✓ INTRODUCTION

This project focuses on analyzing Netflix Movies and TV Shows dataset to understand content trends, distribution patterns, and user preferences. The goal is to extract meaningful insights using data analysis techniques and visualization tools.

✓ OBJECTIVE

The main objectives of this project are:

- To analyze the distribution of Movies and TV Shows
- To identify top content-producing countries
- To study content growth over the years
- To explore popular genres on Netflix

✓ TOOLS USED

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

✓ DATASET DESCRIPTION

The dataset used in this project is the Netflix Movies and TV Shows dataset obtained from Kaggle. It contains information such as title, type, director, cast, country, release year, rating, duration, genre, and description.

IMPORT LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

LOAD DATA

```
import pandas as pd

df = pd.read_excel("netflix_titles.csv.xlsx")
df.head()
```

	show_id	type	title	director	cast	country	date_added	release
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	
			tailbirds					

Next steps:

[Generate code with df](#)[New interactive sheet](#)

DATA UNDERSTANDING

The dataset consists of multiple columns including show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, and description.

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8809 entries, 0 to 8808
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   show_id         8809 non-null   object
 1   type            8809 non-null   object
 2   title           8809 non-null   object
 3   director        8809 non-null   object
 4   cast            8809 non-null   object
 5   country         8809 non-null   object
 6   date_added      8809 non-null   object
 7   release_year    8809 non-null   int64
 8   rating          8809 non-null   object
 9   duration        8809 non-null   object
10   listed_in       8809 non-null   object
11   description     8809 non-null   object
dtypes: int64(1), object(11)
memory usage: 826.0+ KB
```

	release_year
count	8809.000000
mean	2014.181292
std	8.818932
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2024.000000



✓ DATA CLEANING

Data cleaning is performed to handle missing values and remove duplicate records to ensure data quality.

```
df.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

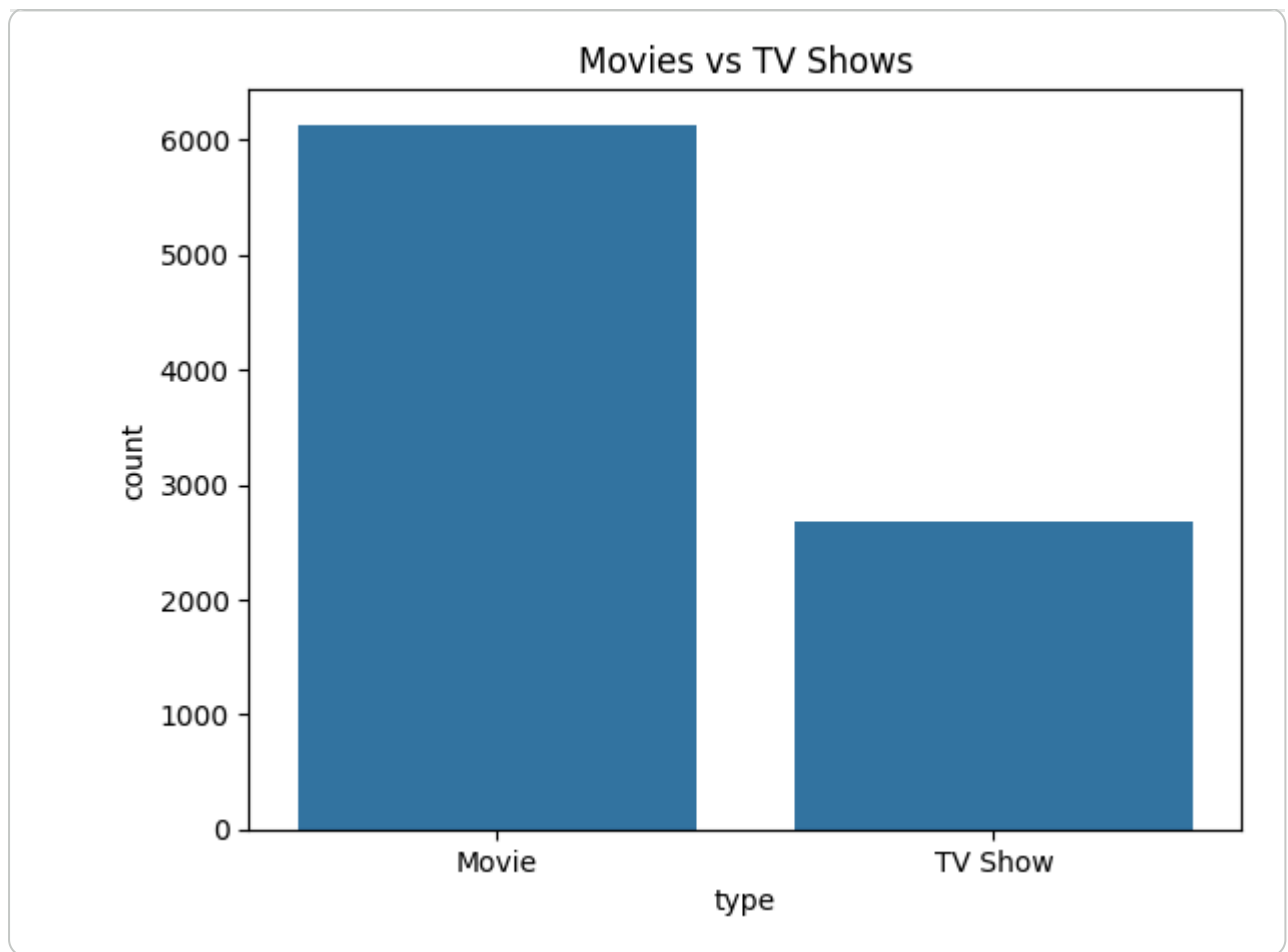
```
df.fillna("Unknown", inplace=True)
df.drop_duplicates(inplace=True)
```

✓ EXPLORATORY DATA ANALYSIS

Movies vs TV Shows

```
import seaborn as sns
import matplotlib.pyplot as plt

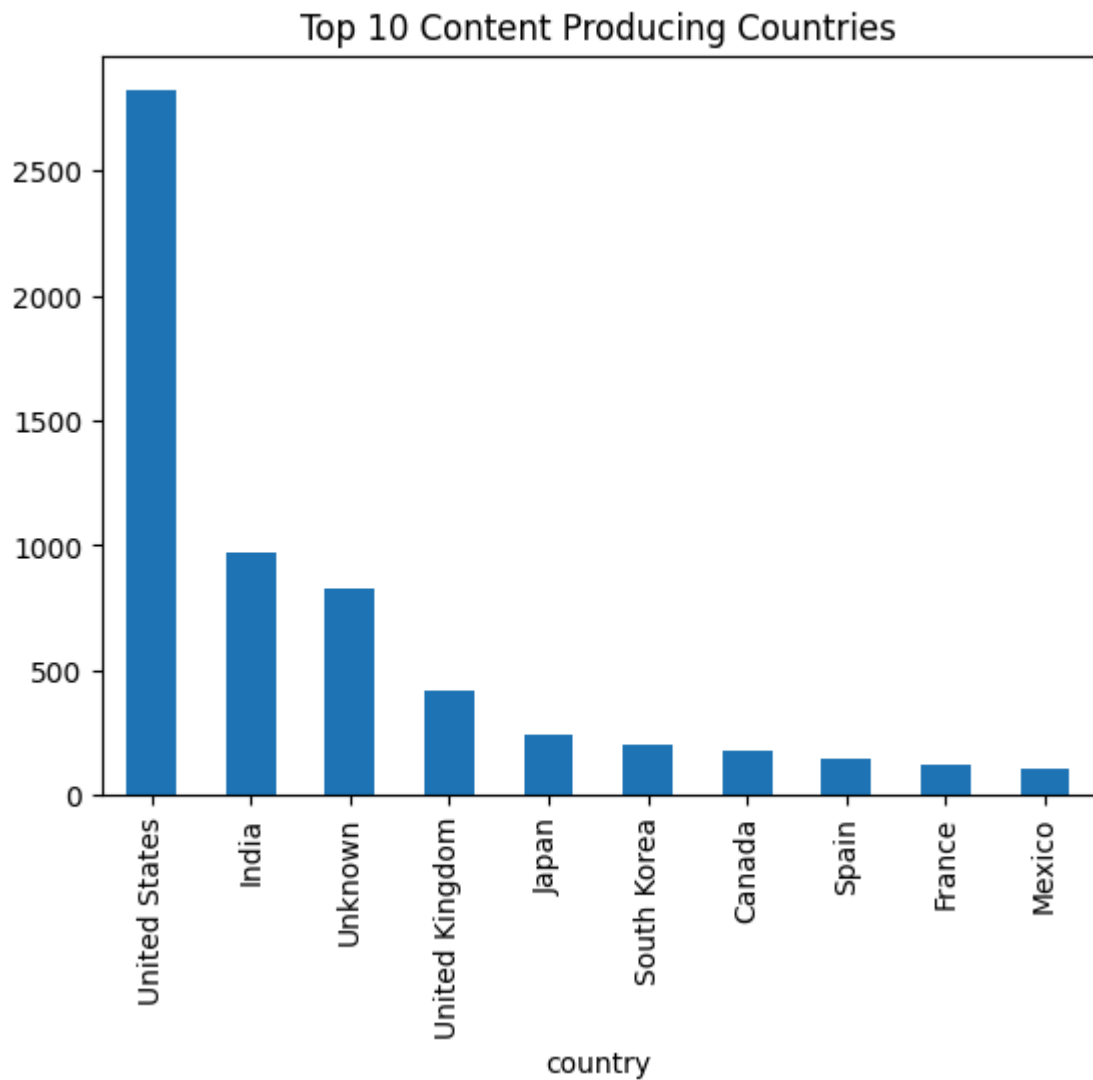
sns.countplot(x='type', data=df)
plt.title("Movies vs TV Shows")
plt.show()
```



This graph shows the distribution of Movies and TV Shows available on Netflix.

Top Countries

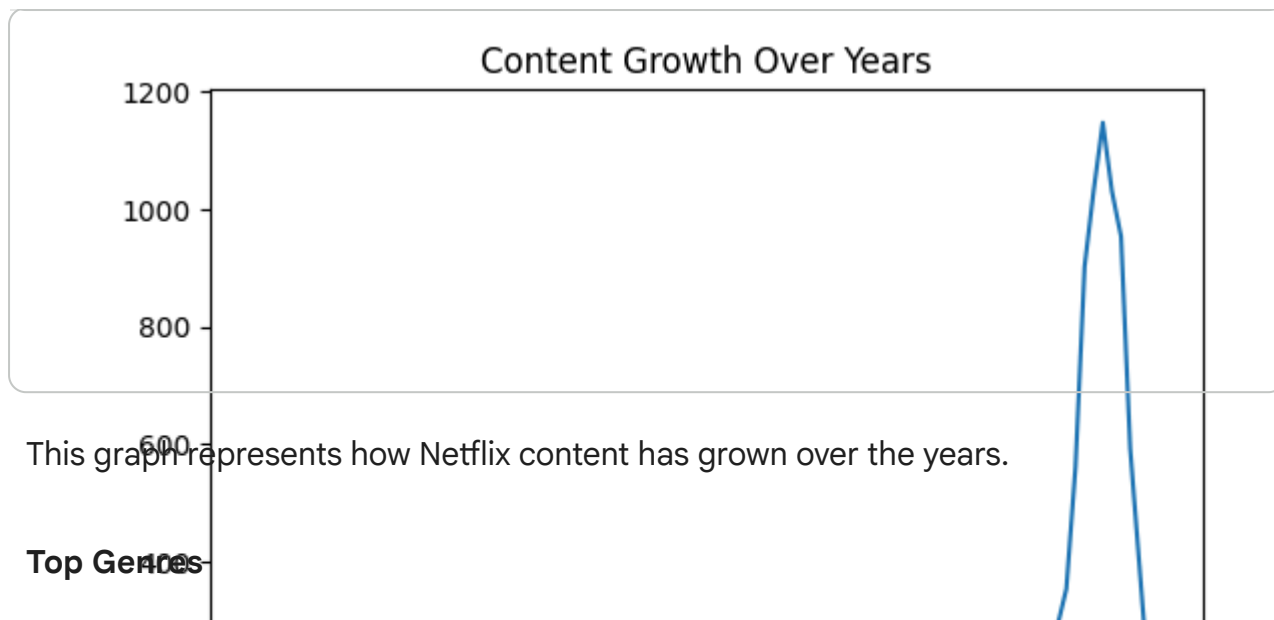
```
df['country'].value_counts().head(10).plot(kind='bar')  
plt.title("Top 10 Content Producing Countries")  
plt.show()
```



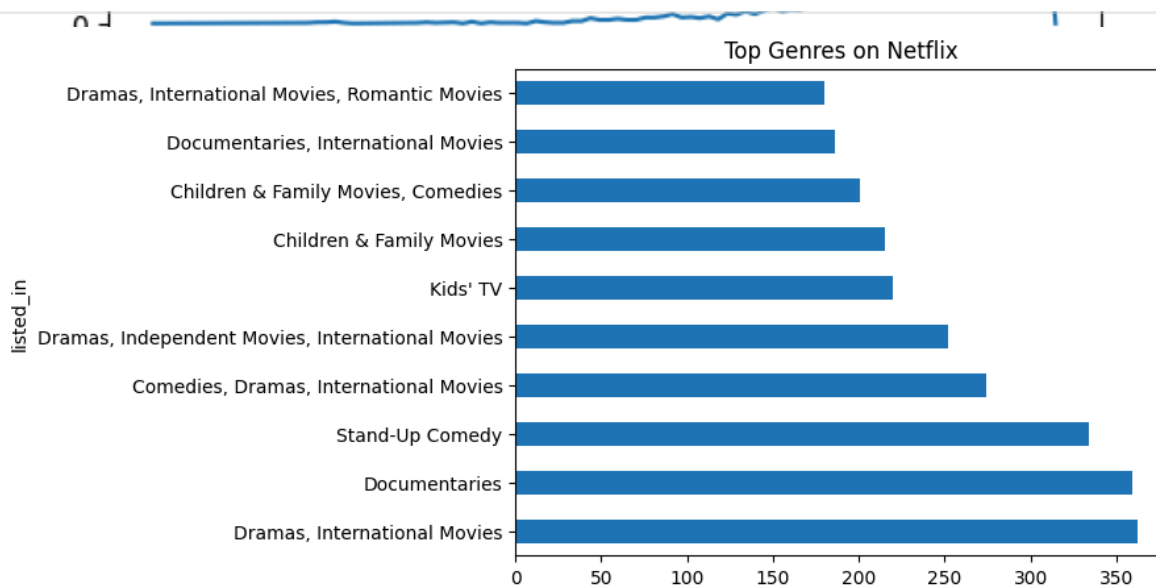
This graph highlights the top countries contributing the most content on Netflix.

Content Growth Over Years

```
df['release_year'].value_counts().sort_index().plot()  
plt.title("Content Growth Over Years")  
plt.show()
```



```
df['listed_in'].value_counts().head(10).plot(kind='barh')
plt.title("Top Genres on Netflix")
plt.show()
```



This graph shows the most popular genres available on Netflix.

✓ KEY INSIGHTS

- Netflix has more Movies compared to TV Shows